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Workpackage D

Smart Energy

Deliverable 2

Implementation of smart meter data in the production of official statistics (actions 6-8)

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Contents

1 Introduction **3**

2 Statistical production processes (GSBPM) **4**

3 Quality framework **6**

4 Metadata **9**

5 Risk plan and mitigation scenarios **10**

6 Data exchange protocols **11**

7 Data storage **13**

8 Data validation **14**
8.1 Data validation during transfer 14
8.2 Data validation during processing 14

9 Data preparation **15**
9.1 Geocoding 15
9.2 Consumption data 16

10 Linking, classification, modelling of consumption **17**

11 Data model of electricity statistics of businesses **19**

Appendices **23**

A When to use big data tools **23**
A.1 Apache Hadoop 23
A.2 PostgreSQL, R and Python 23
A.3 Choosing the tools 24

Abstract

The Smart Energy workpackage (WPD) is one of the workpackages in the ESSnet Big Data II project which aims to implement smart meter data for production of official statistics. This report is the second report out of three. The report gives a brief overview of the statistical products that will be produced by WPD, also a modified statistical production process (GSBPM) is proposed which is adapted for smart meter data. The report further covers micro and macro quality indicators that refer to the input data quality and can be used as a general quality checklist for smart meter data. The report also includes information about metadata and additional information that can be linked from other administrative sources. The report ends with suggestions about different data- exchange protocols, storage solutions, validation rules and preparation methods.

1 Introduction

The use of smart electricity meter data and appropriate analytical methods will enable the ESS to produce new kinds of statistics - trusted smart statistics.

A smart electricity meter measures electricity consumption at a high frequency and communicates the information to a central system. The aim for work package D is to implement smart meter data for producing statistics in different areas, e.g. energy statistics of businesses, census statistics, household costs, tourism seasonality, or impact on environment. The implementation will include linking electricity data with other administrative sources for producing statistics of businesses and households, and identifying vacant living places or seasonal/temporary occupancy of living places. The duration of the project is 24 months and there are four participating countries (Estonia, Denmark, Norway and Sweden). The work is based on the results of the previous ESSnet and results described in previous reports[1;2;3;4] will be used as a starting point to develop functional production prototypes.

It is a great advantage for statistics production if smart meter data are available through a central national hub. Currently, Denmark, Estonia and Norway have national hubs in operation, while Sweden will have a national hub ready by 2021.

The aim of the WP is to develop functional production prototypes from smart meter data. During the project the implementation procedures for the following statistical products will be produced:

1. electricity statistics of businesses, by sector (EE, DK, SE)
2. electricity statistics of households (DK, SE)
3. identifying vacant or seasonally vacant dwellings by new estimation models (EE, NO).

This includes setting up procedures and developing technical solutions to promote and support the collection, processing, and analysis of the data for statistical production.

Additionally, the national hubs will enable the participating NSIs to produce country specific statistical products, for example statistics of finer granularity, new housing statistics, improved statistics on type of production and prepared data available for researchers. Other possible benefits are lower response burden, higher quality, and faster production.

2 Statistical production processes (GSBPM)

To describe the statistical production processes of smart meter data in a general and process-oriented way the Generic Statistical Business Process Model (GSBPM) will be used. GSBPM is the standard adopted for describing production activities performed by NSIs. The principal phases of GSBPM are shown in Figure 1.

Figure 1: Principal phases of GSBPM (Istat 2018)

In order to use GSBPM model in a smart energy context, a small model adaptation is needed. The Italian Statistics Agency ISTAT suggests in their Big Data Committee report to swap the Design and Collect phases, because

only after collection of the data sources, the necessary framework for data processing can be set up. The Design phase can be considered as a Preparation phase, where, rather than doing traditional design choices (e.g. sampling design), a set of preparation activities have to be carried out on the Big Data source[5]. The Italian report is mainly based on experiences with web scraping activities, but the modified production process is of general relevance. The smart energy data are generated for different purposes than official statistical analysis, and they too need to be "prepared". Figure 2 shows the proposed GSBPM that is adapted to Big Data processing and in particular smart energy data processing. The changes mentioned above are highlighted in red.

Figure 2: GSBPM adapted to Big Data processing (Istat 2018)

The Preparation phase for smart energy data processing consists of additional steps, for example

- geocoding of the data
- anonymization of the data

Detailed descriptions of smart energy data processing during the preparation phase are presented in a previous ESSnet Big Data report[1].

In the GSBPM Processing phase the data are checked, cleaned, and transformed so that it can be used for statistical output production. The electricity data are linked and enriched with other administrative data sources that make them suitable for producing statistics of businesses, households and identifying vacant living places.

During the Analyze phase, statistical outputs are produced and examined in detail. Different classification methods are used to identify characteristics of the smart meter data. It includes the mapping between an observed unit - a metering point - and a statistical unit - a business, household or dwelling and assuring good quality outputs from the linked data. The identification of the end consumer can be problematic because consumption measured at a

metering point is assigned to the owner of the contract with an energy company, who is not always the actual consumer. The geocodes of the metering points and exact addresses are important in order to solve this problem. Unsupervised classification algorithms could be used to identify different types of electricity consumption patterns and analyze the possibility to discriminate businesses from private consumers if this information is missing in the smart meter data.

In the Disseminate phase the statistical products are released to the customers. For visualization, clustering and machine learning, tools like Anaconda or Spark can be used.

The evaluation phase manages the evaluation of the statistical business process. It takes place at the end of the process, but relies on inputs gathered throughout the different phases. It includes evaluating the success of a specific instance of the statistical business process, drawing on a range of quantitative and qualitative inputs, and identifying and prioritising potential improvements.

3 Quality framework

A quality framework should ensure the quality of both the processes for creating the statistics, and the resulting output statistics. In the previous ESSnet Big Data, a first list of quality indicators were tested[1;2]. These indicators refer to the quality of the input data and are based on the quality performance indicators developed in the ESSnet on quality of multisource statistics[6]. The purpose is twofold; the indicators can be used for comparisons between countries, and as general quality checklist.

Micro indicators proposed by the ESSnet work package 3 are for example

- Undercoverage
  - Proportion of households and companies that do not have smart meters
  - Proportion of consumption that is not covered
- Percent of meter units that fail basic checks, such as checks for extreme readings
- Percent of meter units where consumption/production level is adjusted

- Percent of meter units where consumption/production level is imputed
- Periodicity, i e frequency of receiving the dataset at the NSI
- Delays in delivery of data
- Percent of units being duplicated in the received dataset
- Differences between administrative and statistical units
- The number of observations per month and per metering point
- Number of days per year and per metering point
- Number of outliers per metering point

Additional macro indicators are

- Comparison of consumption data and the annual consumption of electricity per sector as observed from the survey
- Comparison of consumption data and the output from the household energy consumption survey

At the time, Estonia and Denmark were the only countries with enough hub data available to test the indicators. The development of smartmeter coverage in Denmark is depicted on Figure 3

Attempts to develop a quality framework were also discussed in ESSnet Big Data work package 1 but did not result in any suggested indicators[7]. Two different existing quality frameworks were investigated, the Quality Assessment Framework by Statistics New Zealand (Reid et al 2017)[8] and the Framework for the Quality of Big Data by the UNECE Big Data quality task team (UNECE 2014)[9].

The NZ framework builds on the Total Survey Error (TSE) paradigm (Biemer 2010)[10] and an extension of TSE to administrative and integrated data (Zhang 2012)[11]. It combines the classical statistical thinking of sources of error (i e accuracy) with the notion that modern statistical production involves mixing different data sources. It is not particularly tailored to Big Data sources.

The UNECE framework uses a structure with three hyper dimensions; the source, the metadata, and the data, that apply to the three phases of the

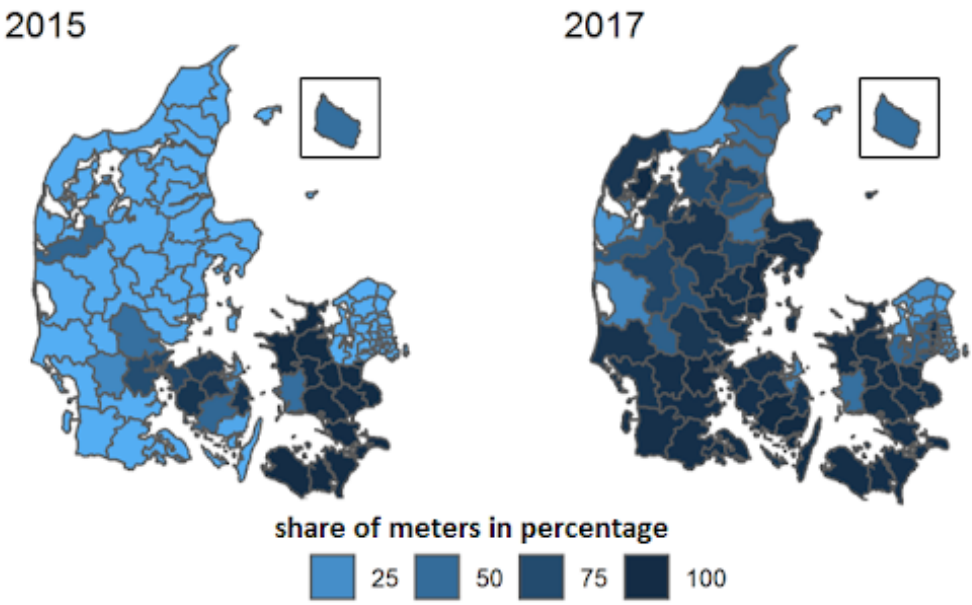

Figure 3: Development of Smartmeter coverage in Denmark

business process; input, throughput, and output. The source relates to the type of data and characteristics of the objects where data are measured, as well as administration and regulations for obtaining data. The metadata refers to descriptions of concepts and content of data sets, and the hyperdimension data relates to the quality of the data itself.

Quality dimensions are nested within the hyperdimensions. Institutional environment, privacy and security are examples of quality dimensions within the source dimension. Traditional quality dimensions such as accuracy and consistency pertains to the data dimension, as does the quality of the linking variables. The metadata dimension describes for example complexity, completeness, usability, linking variables, standardisation, and transparency of methods and processes.

The UNECE framework addresses more quality type issues, compared to the NZ framework. It was concluded in work package 1 that the UNECE framework was more intuitive (for on-line job advertisements) but that the accuracy and selectivity dimensions of the UNECE framework would benefit from the TSE approach of the NZ framework.

The TSE approach has been incorporated in a big data framework by the AAPOR Task Force on Big Data. They introduce the BDTE (Big Data Total Error) framework (AAPOR 2015)[12]. With the BDTE approach, more focus is on errors in the data generating process, compared to the TSE approach.

In their Big Data Committee Annual report for 2017, ISTAT (2018) presents thoughts on a methodology framework for Big Data. They discuss approaches to handle bias introduced by selectivity and validity concerns associated with Big Data sources.

In the present project, we will continue to work on a quality framework for electricity hub data covering not only the input data but also the process of producing statistics, and the statistical output. The work will take into account the results from WP K Methodology and quality.

4 Metadata

Metadata contains information available through the hub, as well as additional information linked from other administrative sources.

Hub data contains information on metering points, customers, agreements and the consumption/production of energy. This could for example be

- Metering point information, including metering point id
- End user information: end user id, street, street number, house number, postal code, postal town/place
- Metering point information: similar to end user information, but in addition the cadastral unit number and the property unit number
- For other metering points: whether metering point for household, cabin, street light, green house etc
- Location coordinates of the metering point

Linking the original data with other data sources (usually administrative registers such as business-, population- and/or of buildings- or dwellings register) gives additional knowledge and enables to produce more varied outputs. The resulting data set including the linked information can be viewed as a statistical register, available within the organisation for use in production of

all kinds of statistics, i e energy statistics as well as economic indicators or environmental statistics.

5 Risk plan and mitigation scenarios

The main risks that can affect the delivery of results for all participating countries in workpackage D can be split into two types of risks:

Data handling risks:

- Protection of personal data fails according to GDPR and other relevant regulations. Electricity consumption data are by many users considered as private, and the perceived breach of privacy increases if the granularity is finer. As an example, electricity consumption may reveal which periods of time that a dwelling is not used. Also, although the inference will be very uncertain, the consumer might be worried about the following: the consumption may suggest some patterns concerning time and frequency of cooking dinner, showering etc. The NSI has access to many administrative registers and have a security policy to ensure compliance with GDPR, and the risk of related to electricity data should not be different than the well-known risks. However, the consequences if the protection should fail, could be much larger due to the perceived risk level among consumers.
- Too many steps the between primary data supplier and utilization of data could make it difficult to fully understand the data
- Linking data to other sources fails or is not of good enough quality
- The data provider changes the data structure
- Poor quality of the outputs

Administrative risks:

- No clear contract with the data provider
- No ongoing contact with the data provider
- Changes in legal aspects opening for the data provider to refuse delivering the data

- Lack of resources and people
- Technological or administrative changes that make previous work obsolete

To meet and mitigate the risks, it is important for the participating countries to set up contracts with the data providers and have good cooperation with them. We must also be able to adapt swiftly to changes in data structure and techniques, and for example investigate the use of models for estimating outputs. Also, we need to make sure that people are trained and that knowledge is transferred, to avoid losing momentum in case of changes in staff.

There are also a number of country specific risks. Norway is expecting to get data of finer granularity, if this is not possible, production might not give the expected results. Denmark had promised to deliver data to researchers, but if there is no benefit for the data provider, they might refuse delivering data. Sweden does not have a hub yet and there is a risk of major changes that will affect the possibility to use the data in the way currently intended. The country specific risks will not be handled by the project.

6 Data exchange protocols

It is suggested to make a contract with the data provider and agree on a fixed delivery frequency and fixed names and datatypes. The obvious and recommend choice of data exchange mean is a secure FTP server. This is doable if one of the parties have an FTP-server available within its business. In most cases that party would be the statistical institute as this is the party interested in receiving data, not only on electrical meters, but from other sources too. When the NSI is the owner and user of the FTP server the data provider logs on to the server and places the data here. The data provider can do this because the NSI has created an account for them to use on the NSI's FTP server.

In order to successfully exchange data, a data provider contract should be in place. In the contract the parties should agree on the following elements in the contract:

- The frequency with which to deliver data and delay. For instance data be delivered every quarter. Data for quarter 1 in 2019 is thus delivered in quarter 2 in 2019.
- Content, datanames, variables, variable names, types, formats.

- The number of files.
  - Typically data is comprised of consumption data and background data. It should be agreed on how many data tables are delivered. Consumption data can be wither hourly, daily, monthly or annually. In some cases a subgroup of the metering points will be read less frequent than the most detailed option. In these cases consumption data, will be delivered in a number of files according to the number of reading options. In the case of Denmark they receive two consumption data tables: one with hourly data and 1 with less frequent readings.
- Is background data a full extract of data for every delivery or will it only be so called deltafiles. Deltafiles here being files with new information.
- A possible format is to exchange zipped csv files. In this case it should be agreed which zip method to use? For instance compression level 9 or 8.
- Character set: UTF-8 or latin or other types.
- Which variables should be delivered in the background file(s)?
- How should the readings in the consumptions dataset(s) be interpreted? Are they ultimo or primo readings? Is it possible to receive a from-date and a to-date also in the consumptions dataset(s)?
- Do the data provider deliver only consumption meters data or also production and exchange meters data. Is a metering point the same as one meter or a group of meters?.

A possible scenario is delivering zipped CSV files to a secure FTP server. For example Denmark has agreed with their dataprovider that files are delivered in compression level 8 and that the character set to be used is UTF-8. It is advisable that it be stated very detailed in the contract how data is delivered. The contract should be formed with assistance from people from the IT department of the NSI and the dataprovider.

7 Data storage

We have investigated whether to use a relational database management system (RDBMS) or Hadoop (HDFS) for data storage.

Smart-meter data in relational databases is feasible and works well, but requires that you adopt an append-only lifecycle of data. All data is appended to existing tables. PostgreSQL (from now on referred to by its common abbreviation Postgres) and Oracle Database are both mature relational database systems (RDBMS) that could be used for data storage. They both feature:

- Extensive support for standard SQL.
- Features relevant for large amounts of data, including table partitioning and parallel query.
- They support a wide range of client connections, including R, Python, .Net, Java and more.
- They are multi-platform, run on Linux, Windows and varying other operating systems.

The key difference between the two databases are licensing fees. Postgres is open-source licensed by an entirely free license: use it as you please, no strings attached, whereas Oracle Database has a very expensive proprietary license. Also parallel query is only available in Oracle Enterprise Edition.

Hadoop by itself is not a database, but a collection of open-source software that runs as a distributed storage framework (HDFS) to manage very large data sets. Its primary purpose is the storage, management, and delivery of data for analytical purposes. At the core Hadoop is a filesystem (Hadoop Distributed File System) that needs additional software (for example, Impala, YARN, Parquet, Hive or Spark) to analyze smart meter data.

Difference between Postgres, Oracle and Hadoop or any other relational database does not necessarily prove that one is better than the other. They all have their strong sides and weaknesses. Choosing the tools for data storage is strongly dependent on data size and the needs and requirements of the NSI. A case study with a more detailed overview and suggestions which technology to use can be found in the appendix A.

8 Data validation

Data validation is a crucial step for detecting outliers and identifying possible errors in the data set. Because of the vast amount of smart meter data, the algorithms used for data validation need to be fast and efficient. There are several steps in the process where it will be meaningful to validate the data. Starting from data delivery to the NSI up until when the final output is published. In this section data validation is split into two major steps: validation during data transaction and data processing.

8.1 Data validation during transfer

Data irregularities that are taken care of as close to data reception as possible is easier corrected for possible irregularities. Upon delivery of data the following validation provisions should be implemented. These provisions are characterized by being about validating if the metadata is as it should be. This is more about data formats, data size, than the value of certain variables themselves.

- Do the files fulfill the requirements written in the contract?
- Do the filenames follow what was agreed upon in the contract?
- Do the variables have the expected formats?
- Are there any new variables?
- Is the number of rows changing drastically from delivery to delivery?
- Are there duplicate rows?

8.2 Data validation during processing

In this step when we start working with data it is possible to detect data irregularities related to the value of certain variables.

- Are there unusual fluctuations occurring in the timeseries. For instance if a household suddenly experience an extreme peak?
- Is it meaningful that a household has a metering point that measures production but not a metering point that measures consumption.
- How many observations contain a zero value? How many observations are null?

- Are metering points from the consumption dataset included in the background dataset and vica versa. Background data being the table that includes information about the adress and the subscription.

9 Data preparation

After data has been collected, some data wrangling has to be done before it can be put to use. Data is typically delivered in at least 3 files. This at least is preferrable. If data is not delivered in this way, it is advised the statistical office split the data themselves to fit this model.

1. Consumption data
2. Address data either the address of the meter or the address of the consumer
3. Subscriptiondata on the meter. Is it a consumption or a production meter, how is it settled etc.

9.1 Geocoding

From the address data it is possible to extract the variables that are related to forming an address: Name of the street, building number, floor number, postal code etc. These are used to forming an address code which then are linked to a coordinate making it insensitive to changes in the street name over time. The coordinates can then be used to link the meter to the following:

- A piece of land
- A building
- A unit within a building

If a metering point can be linked to a unit within a building, then it means that it is also linked to a building and a piece of land. An example from Denmark is depicted in Figure 4.

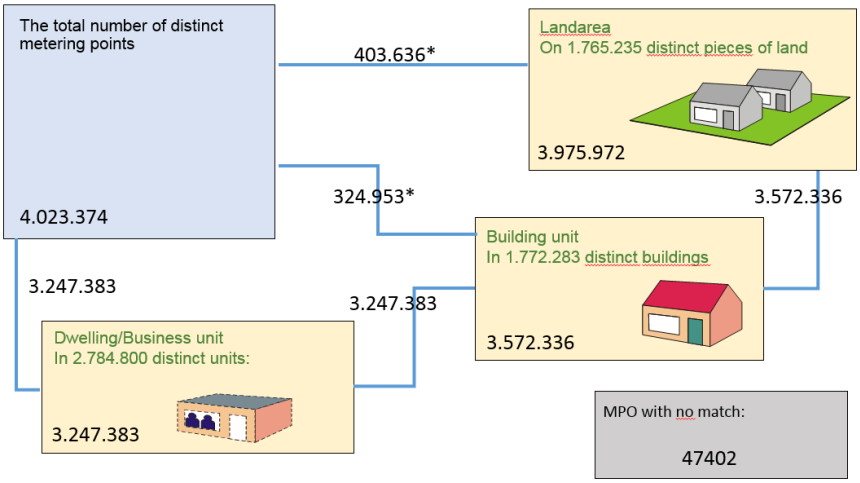

Figure 4: Linking the meters to geographical places in the form of land area, buildings or units.

There are 4.023.374 distinct metering points, they can be divided into points that can both be linked to:

- Dwelling/Buisness units, building, land area: 3.247.383
- Buildings and land area: $3.247.383 + 324.953 = 3.572.336$
- Landarea: $3.247.383 + 324.952 + 403.636 = 3.975.972$
- No match: 47.402

9.2 Consumption data

Consumption data needs to be handled according to the desired outputs. Mainly the desired outputs are aggregated statistics by sector. This be households or businesses. In this step of the collection the statistical office must set up a system that brings data into the desired aggregation level. For instance monthly aggregated data by metering point.

10 Linking, classification, modelling of consumption

In all the participating countries, administrative datasets - business register, population register, and/or register of buildings or dwellings - are used for linking. One important reason for linking to business information is to capture the NACE (Nomenclature of Economic Activities) code for local units from where it is possible to get the consumption by economic activity. The linking can be done either by business code or by the address, possibly aided by coordinates. From the population register, people living at the same address can be identified as a household, and the data can be linked with electricity data by address or a unique dwelling identification.

In principle linking two data sets should be easy. There is a common unique identifier that can be used to join two or more tables. The unique identifier can be a number or code that uniquely identifies a person, business or organisation registry code, address id or some other unique value that is used in the NSI's. Problems arise when the quality of the keys is poor and different approaches and strategies must be combined to increase the linking quality.

To find the electricity statistics of businesses by sector, monthly aggregated data is sufficient. The exact procedure for linking depends on the data at hand but general guidelines for linking are:

1. Finding all the businesses from metering data.
2. Filtering out businesses with a non valid grid agreement.
3. Filtering out boarderpoints (metering points between two network operators).
4. JJoin the businesses by address id or building id or unit id with statistical business register.
5. Filter out buildings and unit codes which belong to apartment associations.

Depending on the quality of linking there might be problems where on the same address are several metering points or several statistical units (companies). Finding the actual end consumer and identifying the amount of electricity consumed or produced might be problematic. Two different strategies are tested to solve the problem. Firstly the consumption is divided evenly between companies that share the same address and secondly the consumption

is weighted by the number of employees in the company. Also the economic activity of a business can be taken into account. Which strategy to choose depends strongly on the data set and administrative registers that are available in the NSI. To compute the annual and quarterly electricity consumption statistics by economic activity sector the NACE code is used to group each business into an economic sector and the consumption data is aggregated. The survey estimates can be compared with the smart meter electricity data estimates and if need be the strategies of how to divide the consumption between entities can be adjusted.

11 Data model of electricity statistics of businesses

The flow of smart meter data can be divided into three larger groups. Data receival and preparation phase, where the NSI's receive the data from the datahub/provider. This phase includes preliminary data validation, anonymization of the data (personal codes, customer codes etc..) and geocoding of the data. A typical data model of this phase can be seen on Figure 5.

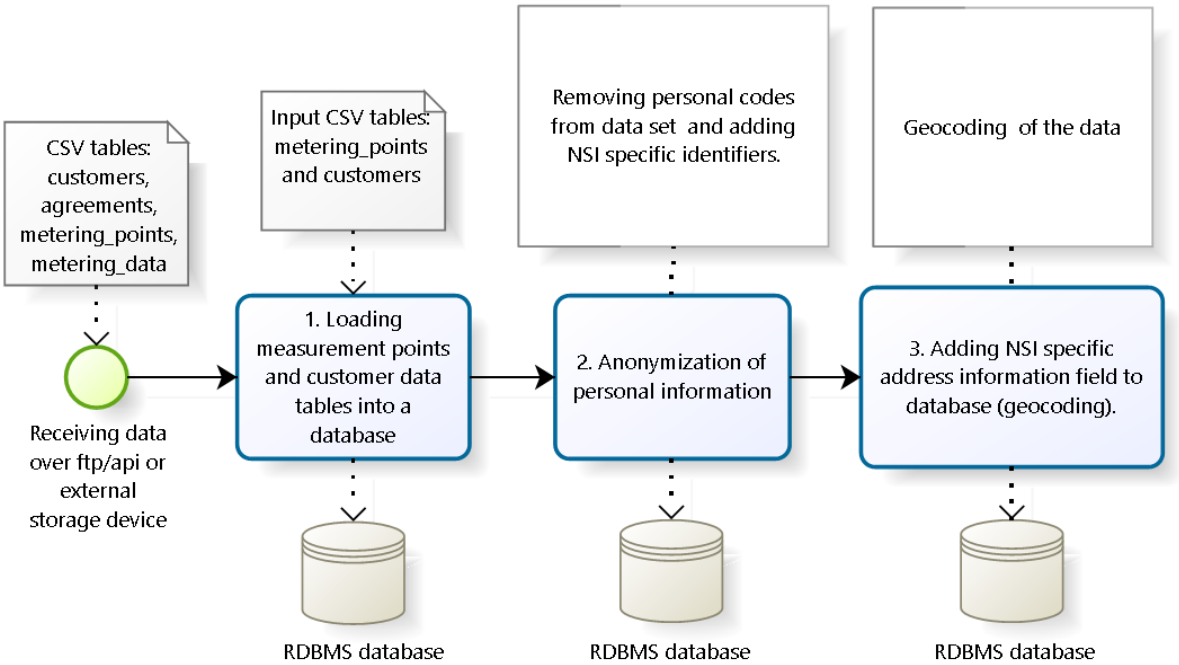

Figure 5: Data receival and preparation phase

The second stage is storing and aggregating the data (Figure 6). The data can be stored in a classical RDBMS or a more big data oriented solution like Hadoop HDFS. For most use cases the data granularity doesn't have to be very fine and daily or even monthly aggregates will save computing power and time.

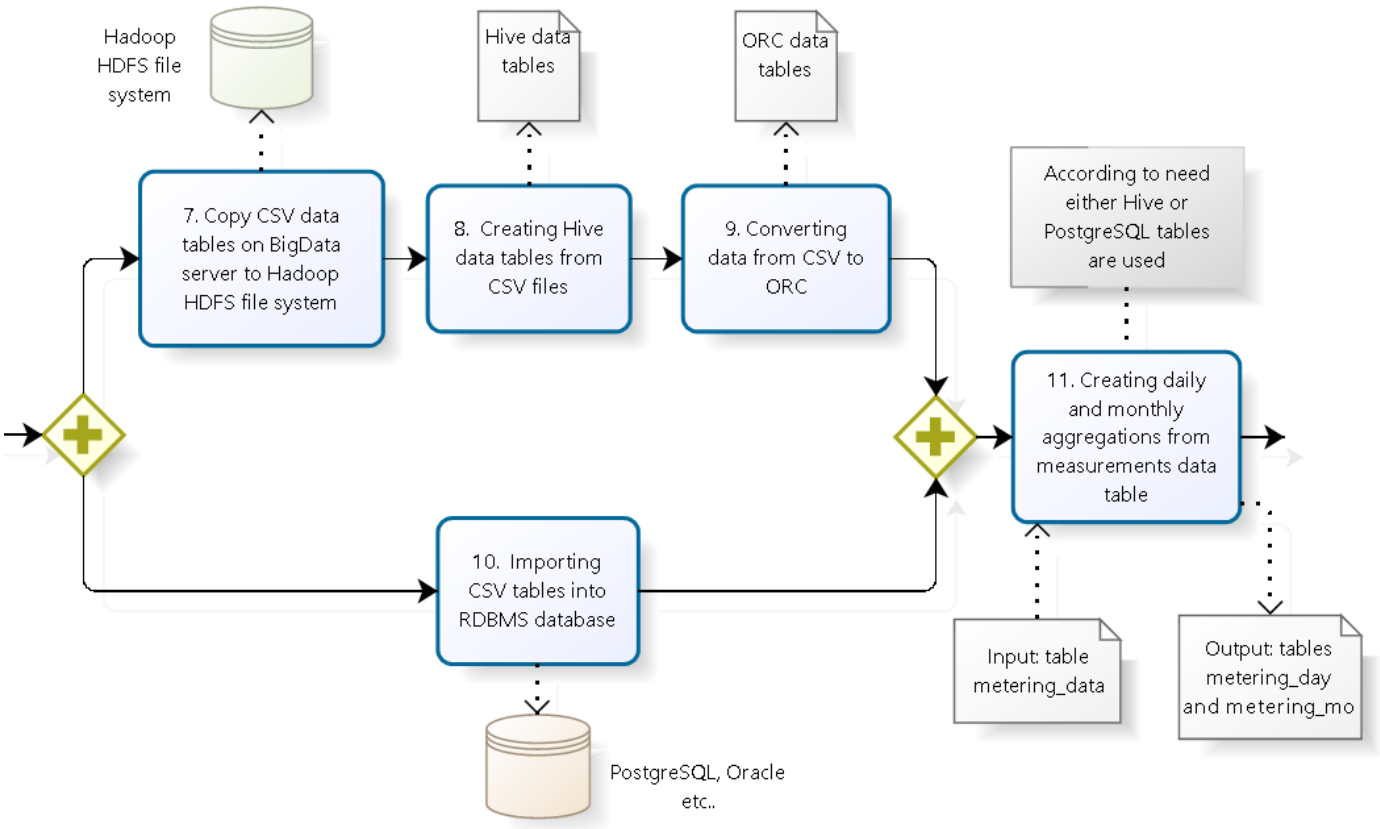

Figure 6: Schematic for storing and aggrgating smart meter data

The third phase of the process is linking the electricity data to administrative registers and validating the outputs (Figure 7). Linking can enhance the data quality and it gives us the possibility to calculate the electricity consumption statistics by economic activity sector.

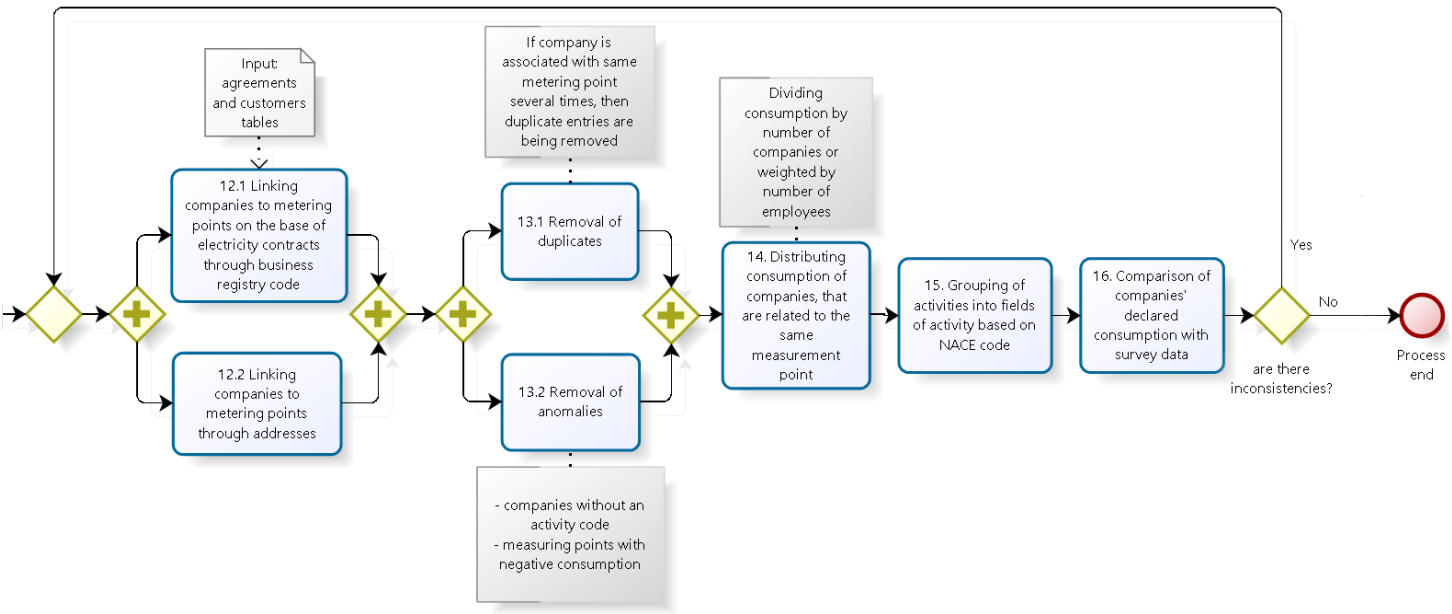

Figure 7: Data linking and validation phase

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A When to use big data tools

In Estonia we have evaluated two solutions for data storage and processing:

1. Apache Hadoop framework both as a data storage backend (HDFS) and data querying and processing platform (Hive and Spark).
2. PostgreSQL database together with widely used data science tools available for R and Python languages.

A.1 Apache Hadoop

The Apache Hadoop software library is a framework that allows for the distributed processing of large data sets across clusters of computers using simple programming models. It is designed to scale up from single servers to thousands of machines, each offering local computation and storage. Rather than rely on hardware to deliver high-availability, the library itself is designed to detect and handle failures at the application layer, so delivering a highly-available service on top of a cluster of computers, each of which may be prone to failures.

Hadoop Distributed File System (HDFS) is a distributed file system that provides high-throughput access to application data.

Hive is a data warehouse infrastructure that provides data summarization and ad hoc querying.

Spark is a fast and general compute engine for Hadoop data. Spark provides a simple and expressive programming model that supports a wide range of applications, including ETL, machine learning, stream processing, and graph computation.

A.2 PostgreSQL, R and Python

PostgreSQL is a powerful, open source object-relational database system that uses and extends the SQL language combined with many features that safely store and scale the most complicated data workloads.

R is a programming language and free software environment for statistical computing and graphics. R has grown from a statistic package to a popular data science toolset with a thriving community, lots of add-on packages and

visualisation options.

Python is a high-level, general-purpose programming language. One of the most popular programming languages in the world, Python has also extensive set of data science packages.

A.3 Choosing the tools

Probably the most important aspect while choosing the tools is to look at the dataset size. In Estonia we have hourly data from all metering points, which sums up to about 6.5 billion records per year (Table 1).

Timeframe	**Number of records**
One hour	750,000
One day	18,000,000
One month (approximately)	540,000,000
One year	6,570,000,000

Table 1: Hourly smart meter records and number of rows

In official statistics hourly and even daily data is rarely needed, so to optimise further processing we calculate aggregates for every day, month and year and store it along with the raw hourly data.

Aggregates	**Number of records**
365 days	273,750,000
12 months	9,000,000
One year	750,000

Table 2: Aggregated smart meter data and the number of rows they

Using the aggregate datasets (Table 2) means that raw data almost never needs to be accessed and resource requirements for memory and processing power are considerably smaller. In our case, due to the relatively low number of metering points, we have found that using PostgreSQL for data storage is better solution compared to the Hadoop. On the same hardware response times for most of the queries are up to ten times worse with Hadoop tools.

Another thing to consider is the complexity of installing and managing Apache Hadoop cluster - this requires specific skills that might not be readily available in your organisation. Also with small scale setup more hardware resources are needed for Hadoop compared to the use of simple relational database. Apache Hadoop should be considered in cases when

- data size is much larger (more than ten million metering points with hourly data)
- highly parallelizable algorithms are used that can benefit from the distributed processing